

390. The synthetic rubber has replaced natural rubber for domestic and industrial purposes. Which one of the following is the main reason behind that? **2016 I**

- (a) Natural rubber is unable to meet the growing demand of different industries.
- (b) Natural rubber is grown in tropical countries only
- (c) Raw materials for synthetic rubber are easily available
- (d) Natural rubber is not durable

391. The handle of pressure cookers is made of plastic because it should be made non-conductor of heat. The plastic used there is the first man-made plastic, which is **2016 I**

- (a) polythene
- (b) terylene
- (c) nylon
- (d) bakelite

Environment and its Pollution

392. How much amount of sun rays reaching the earth is used by plants?
(a) 90% (b) 1% (c) 15% (d) 48%

393. The pollutants which come directly in the air from sources are called primary pollutants. Primary pollutants are sometimes converted into secondary pollutants. Which of the following belongs to secondary air pollutants?
(a) CO
(b) Hydrocarbon
(c) Peroxyacetyl nitrate (PAN)
(d) NO

394. Ozone is formed in the upper atmosphere from oxygen by the action of
(a) thermal radiation from sunlight
(b) cosmic rays
(c) ultraviolet rays
(d) infrared rays

395. Which of the following statements is wrong?
(a) Ozone is not responsible for greenhouse effect
(b) Ozone can oxidise sulphur dioxide present in the atmosphere to sulphur trioxide
(c) Ozone hole is thinning of ozone layer present in stratosphere
(d) Ozone is produced in upper stratosphere by the action of UV rays on oxygen

396. Which one of the following is the major cause of depletion of ozone strata of the atmosphere?

- (a) Carbon dioxide
- (b) Chlorofluoro carbon
- (c) Sulphur dioxide
- (d) Nitrogen oxide

397. Which of the following statements is correct?

- (a) Ozone hole is a hole formed in stratosphere from which ozone ooze out
- (b) Ozone hole is a hole formed in the troposphere from which ozone ooze out
- (c) Ozone hole is thinning of ozone layer of stratosphere at some places
- (d) Ozone hole means vanishing of ozone layer around the earth completely

398. Oxidation of sulphur dioxide into sulphur trioxide in the absence of a catalyst is a slow process but this oxidation occurs easily in the atmosphere. Which substance catalyse the reaction?

- (a) Oxygen
- (b) Particulate
- (c) UV rays
- (d) IR rays

399. Dinitrogen and dioxygen are main constituents of air but these do not react with each other to form oxides of nitrogen because

- (a) the reaction is endothermic and requires very high temperature
- (b) the reaction can be initiated only in the presence of a catalyst
- (c) oxides of nitrogen are unstable
- (d) N₂ and O₂ are unreactive

400. In acid rain, the acid present in highest concentration is

- (a) nitric acid
- (b) hydrochloric acid
- (c) sulphuric acid
- (d) carbonic acid

401. Which of the following statements is not true?

- (a) London smog is a mixture of smoke and fog
- (b) London smog is oxidising in nature
- (c) Photochemical smog causes irritation in eyes
- (d) Photochemical smog results in the formation of PAN

402. Which of the following statements is not true about classical smog?

- (a) Its main components are produced by the action of sunlight on emissions of automobiles and factories
- (b) Produced in cold and humid climate

- (c) It contains compounds of reducing nature
- (d) It contains smoke, fog and sulphur dioxide

403. Which of the following compounds caused tragedy of Bhopal in 1984?

- (a) Phosphene
- (b) Methyl isocyanate
- (c) Carbon monoxide
- (d) Methyl cyanate

404. SO₂ and NO₂ causes pollution by increasing

- (a) alkalinity
- (b) neutrality
- (c) acidity
- (d) buffer action

405. Freon used as refrigerants is chemically known as

- (a) chlorinated hydrocarbon
- (b) fluorinated hydrocarbon
- (c) chlorofluoro hydrocarbon
- (d) fluorinated aromatic compound

406. Which one of the following is considered as point source for water pollution?

- (a) Factories
- (b) Construction site
- (c) Acid rain
- (d) Field lawns

407. Sewage containing organic waste should not be disposed in water bodies because it causes major water pollution. Fishes in such a contaminated water die because of

- (a) large number of mosquitoes
- (b) increase in the amount of dissolved oxygen
- (c) decrease in the amount of dissolved oxygen
- (d) clogging of gills by mud

408. Biochemical Oxygen Demand (BOD) is a measure of organic materials present in water. BOD value less than 5 ppm indicates a water sample to be

- (a) rich in dissolved oxygen
- (b) poor in dissolved oxygen
- (c) highly polluted
- (d) not suitable for aquatic life

409. For dry cleaning, in the place of tetrachloroethane, liquefied carbon dioxide with suitable detergent is an alternative solvent. What type of harm to the environment will be prevented by stopping use of tetrachloroethane?

- (a) It results in tropospheric pollution
- (b) It causes depletion of ozone layer
- (c) It causes particulate pollution
- (d) Both (a) and (b)

410. From which of the following water sources, the water is likely to be contaminated with fluoride?

1. Ground water
2. River water

Choose the correct code

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

411. Consider the following statements.

1. Ultraviolet rays coming from sun causes skin cancer.
2. Taj Mahal is threatened by pollution from SO₂

Choose the correct statement and select the correct code.

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

412. Statement I Oxides of sulphur and nitrogen present in high concentration in air are dissolved in rain drop.

Statement II Oxy acids of sulphur and nitrogen makes rain water acidic.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
- (b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
- (c) Statement I is correct but Statement II is incorrect
- (d) Statement I is incorrect but Statement II is correct

413. Which of the following occupational health hazards commonly faced by the workers of ceramics, pottery and glass industry.

1. Stone formation in gall bladder
2. melanoma
3. silicosis

Choose the correct code

- (a) Only 2
- (b) Only 3
- (c) 1 and 2
- (d) All of the above

414. Ordinary dry air consists of the following.

1. Nitrogen
2. Argon
3. Oxygen
4. Carbon dioxide

What is the decreasing sequence of these in percentages?

- (a) 1, 3, 2 and 4
- (b) 1, 2, 4 and 3
- (c) 2, 1, 3 and 4
- (d) 2, 1, 4 and 3

415. Match the following Columns.

Column I (Pollutant)	Column II (Source)
A. Microorganisms	1. Chemical fertilisers
B. Plant nutrients	2. Abandoned coal mines
C. Sediments	3. Domestic sewage
D. Mineral acids	4. Erosion of soil by strip mining
	5. Detergents

Codes

- | | |
|-------------|-------------|
| A B C D | A B C D |
| (a) 2 5 3 1 | (b) 3 1 4 2 |
| (c) 4 2 5 1 | (d) 1 3 2 4 |

416. Which one of the following is associated with the formation of brown air in traffic congested cities?

- (a) Sulphur dioxide
- (b) Nitrogen oxide
- (c) Carbon dioxide
- (d) Carbon monoxide

417. Which one of the following chemicals is commonly used by farmers to destroy weeds?

- (a) DDT
- (b) Malathion
- (c) Methyl bromide
- (d) 2,4-D

418. From which one among the following water sources, the water is likely to be contaminated with fluoride?

- (a) Ground water
- (b) River water
- (c) Pond water
- (d) Rain water

419. Which of the occupational health hazards commonly faced by the workers of ceramics, pottery and glass industry is

- (a) stone formation in gall bladder
- (b) melanoma
- (c) silicosis
- (d) stone formation in kidney

> Previous Years' Questions

420. Which one among the following is responsible for the expansion of water in the ocean? **2013 I**

- (a) Carbon dioxide
- (b) Nitrogen dioxide
- (c) Carbon monoxide
- (d) Sulphur dioxide

421. Which one among the following gases readily combines with the haemoglobin of the blood? **2013 I**

- (a) Methane
- (b) Nitrogen dioxide
- (c) Carbon monoxide
- (d) Sulphur dioxide

422. Which of the following gases in the atmosphere is/are responsible for acid rain? **2013 I**

1. Oxides of sulphur
2. Oxides of nitrogen
3. Oxides of carbon

Select the correct answer using the codes given below.

- (a) 1 and 2
- (b) 1 and 3
- (c) Only 2
- (d) All of the above

423. Match the following Columns.

2013 I

Column I (Agent)	Column II (Disease)
A. Arsenic	1. Fluorosis
B. Fluoride	2. Melanosis
C. Dust	3. Presbycusis
D. Noise	4. Silicosis

Codes

- | |
|-------------|
| A B C D |
| (a) 3 1 4 2 |
| (b) 3 4 1 2 |
| (c) 2 1 4 3 |
| (d) 2 4 1 3 |

424. Statement I Chlorine radicals Cl[•] initiate the chain reaction for ozone depletion.

Statement II Gaseous hypochlorous acid and chlorine are photolysed by sunlight.

Codes **2013 I**

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
- (b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
- (c) Statement I is correct but Statement II is incorrect
- (d) Statement I is incorrect but Statement II is correct

425. Nitric oxide pollution can lead to all of the following, except

2014 I

- (a) leaf spotting in plants
- (b) bronchitis related respiratory problems in human
- (c) production of corrosive gases through photochemical reaction
- (d) silicosis in human

426. The technique of inducing rain from cloud is called? **2014 II**

- (a) Cloud computing
- (b) Cloud control
- (c) Cloud engineering
- (d) Cloud seeding

427. Match the following Columns.

☞ 2015 I

Column I (Air pollutants)	Column II (Effect)
A. Chlorofluorocarbon	1. Acid rain
B. Sulphur dioxide	2. Depletion in ozone layer in the atmosphere
C. Lead compound	3. Harmful for human nervous system
D. Carbon dioxide	4. Topmost contribution to greenhouse effect.

Codes

A B C D	A B C D
(a) 4 3 1 2	(b) 4 1 3 2
(c) 2 1 3 4	(d) 2 3 1 4

428. Which one of the following is the most appropriate and correct practice from the point of view of a healthy environment? ☞ 2015 I

- Burning of plastic wastes to keep the environment clean
- Burning of dry and fallen leaves in a garden or field
- Treatment of domestic sewage before its release
- Use of chemical fertilisers in agricultural fields

429. Which of the following statement(s) is/are correct?

- Acid rain reacts with buildings made from limestone.
- Burning of sulphur containing coal can contribute to acid rain.

3 Eutrophication is an effective measure to control pollution.

Select the correct answer using the codes given below. ☞ 2016 I

- 1 and 2
- 2 and 3
- Only 1
- All of the above

430. Methyl isocyanate gas, which was involved in the disaster in Bhopal in December, 1984, was used in the Union Carbide factory for production of ☞ 2016 I

- dyes
- detergents
- explosives
- pesticides

> ANSWERS

1	a	2	c	3	b	4	b	5	a	6	c	7	b	8	a	9	b	10	a
11	b	12	c	13	c	14	b	15	a	16	c	17	d	18	a	19	a	20	a
21	c	22	d	23	c	24	d	25	d	26	d	27	d	28	b	29	b	30	b
31	d	32	d	33	c	34	b	35	d	36	b	37	b	38	c	39	a	40	d
41	a	42	d	43	b	44	c	45	a	46	a	47	d	48	a	49	d	50	a
51	a	52	c	53	d	54	a	55	b	56	b	57	c	58	b	59	c	60	b
61	b	62	b	63	b	64	b	65	b	66	c	67	b	68	b	69	c	70	a
71	c	72	c	73	a	74	d	75	b	76	a	77	c	78	d	79	c	80	b
81	c	82	a	83	b	84	b	85	d	86	c	87	d	88	d	89	b	90	a
91	b	92	d	93	c	94	c	95	b	96	d	97	a	98	c	99	c	100	d
101	a	102	d	103	a	104	c	105	b	106	c	107	b	108	a	109	b	110	d
111	b	112	a	113	c	114	c	115	d	116	b	117	c	118	b	119	a	120	a
121	d	122	a	123	c	124	c	125	d	126	a	127	c	128	a	129	d	130	c
131	a	132	c	133	b	134	a	135	c	136	c	137	a	138	c	139	b	140	d
141	c	142	c	143	d	144	d	145	a	146	b	147	b	148	b	149	a	150	c
151	d	152	b	153	a	154	c	155	b	156	b	157	c	158	a	159	a	160	a
161	c	162	a	163	a	164	d	165	d	166	c	167	b	168	b	169	d	170	a
171	d	172	b	173	a	174	d	175	b	176	c	177	b	178	c	179	d	180	a
181	b	182	a	183	a	184	c	185	b	186	a	187	b	188	b	189	b	190	c
191	b	192	d	193	d	194	a	195	d	196	d	197	a	198	c	199	c	200	b
201	a	202	c	203	c	204	d	205	c	206	c	207	b	208	d	209	a	210	c
211	a	212	c	213	d	214	c	215	c	216	b	217	b	218	c	219	c	220	a
221	a	222	a	223	c	224	c	225	a	226	c	227	d	228	b	229	c	230	a
231	a	232	a	233	d	234	c	235	a	236	b	237	b	238	b	239	a	240	d
241	a	242	d	243	c	244	a	245	d	246	b	247	d	248	a	249	d	250	a
251	d	252	d	253	d	254	a	255	d	256	d	257	d	258	c	259	d	260	a
261	a	262	b	263	c	264	a	265	a	266	c	267	c	268	d	269	b	270	b

271	b	272	a	273	a	274	a	275	b	276	c	277	c	278	a	279	b	280	c
281	a	282	b	283	d	284	d	285	a	286	c	287	b	288	a	289	b	290	c
291	c	292	b	293	b	294	d	295	c	296	d	297	d	298	b	299	d	300	c
301	c	302	c	303	a	304	a	305	a	306	c	307	b	308	a	309	d	310	b
311	a	312	d	313	a	314	c	315	a	316	b	317	d	318	d	319	a	320	b
321	c	322	a	323	d	324	d	325	d	326	c	327	c	328	a	329	b	330	b
331	a	332	d	333	a	334	a	335	a	336	a	337	c	338	c	339	a	340	b
341	c	342	b	343	b	344	b	345	d	346	b	347	d	348	c	349	d	350	c
351	b	352	b	353	c	354	c	355	a	356	b	357	c	358	b	359	d	360	c
361	b	362	a	363	d	364	c	365	a	366	a	367	c	368	b	369	d	370	c
371	b	372	a	373	d	374	c	375	c	376	b	377	d	378	d	379	b	380	c
381	a	382	c	383	b	384	c	385	d	386	a	387	b	388	d	389	b	390	c
391	d	392	b	393	c	394	c	395	a	396	b	397	c	398	b	399	a	400	c
401	b	402	a	403	b	404	c	405	c	406	a	407	c	408	a	409	b	410	b
411	c	412	a	413	b	414	a	415	b	416	b	417	d	418	b	419	c	420	a
421	c	422	a	423	c	424	b	425	b	426	d	427	c	428	c	429	a	430	c

Solutions

5. At STP, 44 g (1 mole) carbon dioxide
 $= 6.02 \times 10^{23}$ molecules

7. Density of liquid is generally less than that of its solid form. But solid ice is lighter than water as it floats on water, i.e. the density of solid form of water (ice) is less than that of the liquid form of water.

14. Pure water reaches at its maximum density, when cooled at approximately 4°C (39°F). As it is cooled further, at 0°C, water expands to become less dense.

This unusual negative thermal expansion is attributed to strong, orientation dependent, intermolecular interactions (hydrogen bonds).

16. Chemical changes are generally irreversible. So, the frying of an egg is a chemical change.

17. Molecular weight of NaCl = 23 + 35.5 = 58.5

Equivalent weight

$$= \frac{\text{Molecular weight}}{\text{Valence factor}} = \frac{58.5}{1} = 58.5$$

21. Latent heat is the amount of heat required per mass during the change of phase of a substance.

SI unit J kg^{-1}

$$\text{Latent heat (L)} = \frac{\text{Quantity of heat (Q)}}{\text{Mass (m)}}$$

Here, $m = 1100$ g of ice

$$L = 333.6 \text{ J/g}$$

$$\text{So, amount of heat (Q)} = m \times L = 100 \times 333.6$$

$$Q = 33360 \text{ J}$$

$$109. \frac{r_H}{r_{He}} = \sqrt{\frac{M_{He}}{M_H}} = \sqrt{2}$$

$$r_H = 1.4 r_{He}$$

$$114. \text{Molarity} = \frac{\text{Mass of solute}}{\text{Molar mass of solute} \times \text{volume of solution (L)}} \\ = \frac{8}{40 \times 1} \quad (\because \text{NaOH} = 23 + 16 + 1 = 40 \text{ g mol}^{-1}) \\ = \frac{1}{5} = 0.2 \text{ M}$$

$$115. \text{Molarity of pure water} = \frac{1000}{18} = 55.55 \text{ M}$$

120. The tanks used by scuba divers are filled with air diluted with helium (i.e. He—O₂ mixture) because unlike nitrogen, helium is not soluble in blood even under high pressure.

147. Soft drinks regulate acidity with bicarbonate salts. Bicarbonate of soda is effective in regulating the pH levels of other substances, it ensures that the substance is neither too alkaline nor too acidic.

148. The blue litmus paper turns red in acidic solution and the red litmus paper turns blue in basic solution. The neutral solution does not affect the litmus paper. Thus, only acidic solution can change the colour of blue litmus paper to red.

241. Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is added to clinker during cement manufacturing to decrease the rate of setting of cement so that it gets sufficiently hardens.

242. Pearl consists of approximately 85% calcium carbonate hence, the main constituent of a pearl is calcium carbonate.

- 274.** The hardness of water is mainly due to the presence of Ca or Mg ions. When hard water containing these ions is treated with soap solution, it reacts to form white crudy precipitate, known as scum.

- The fourth valence electron of each carbon atom is 'free' to move, which is responsible for the conduction of electricity. In a diamond crystal, each carbon atom is linked to four other carbon atoms by strong covalent bonds.

